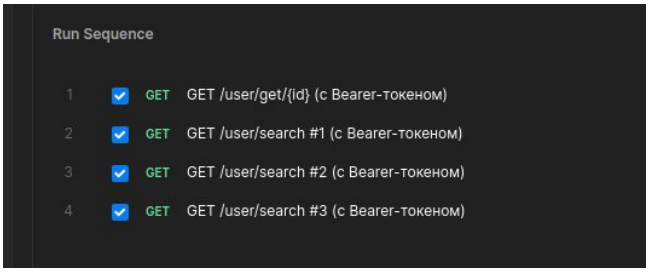
1. План нагрузочного тестирования

1:3

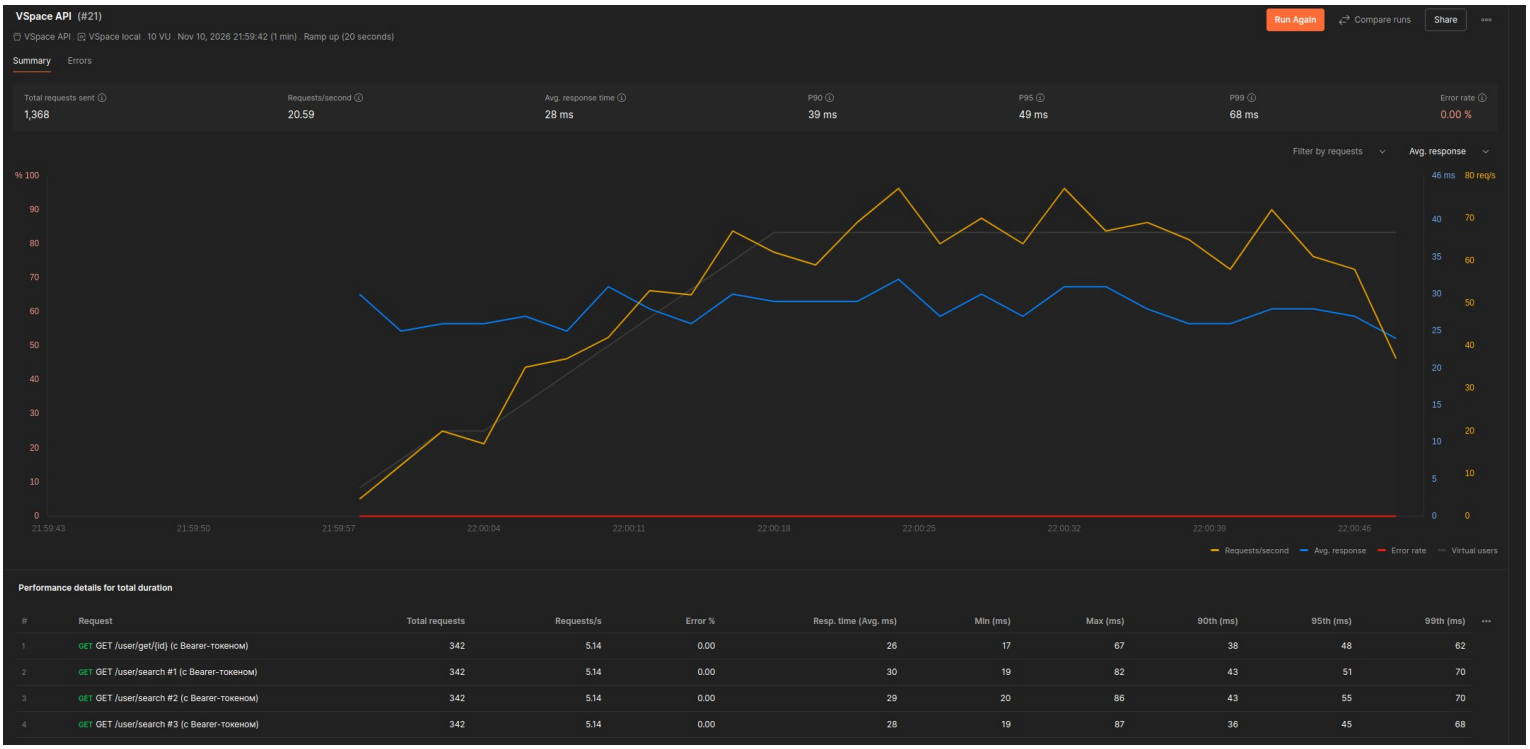
/user/get/{id}

/user/search

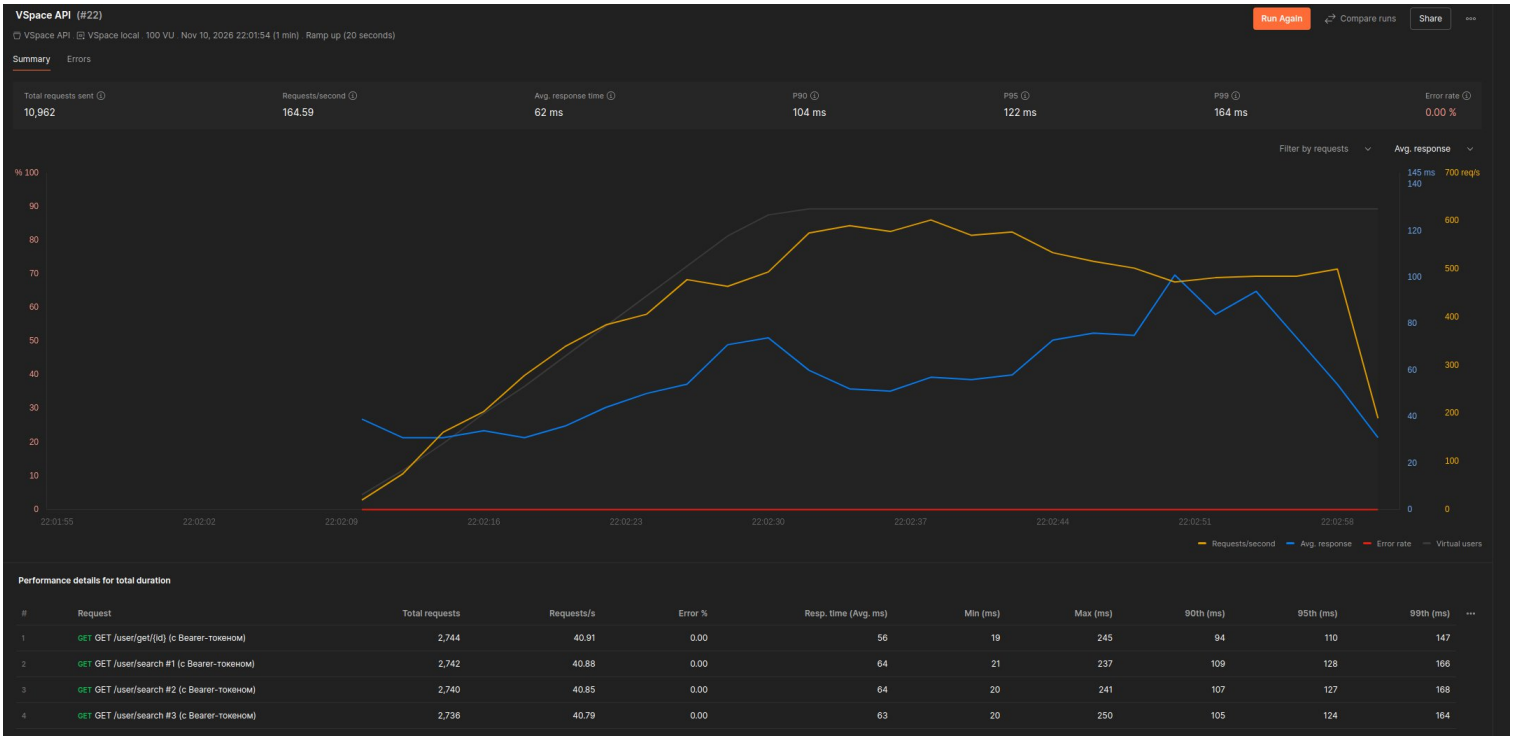


2. Замеры (master only)

10 VU



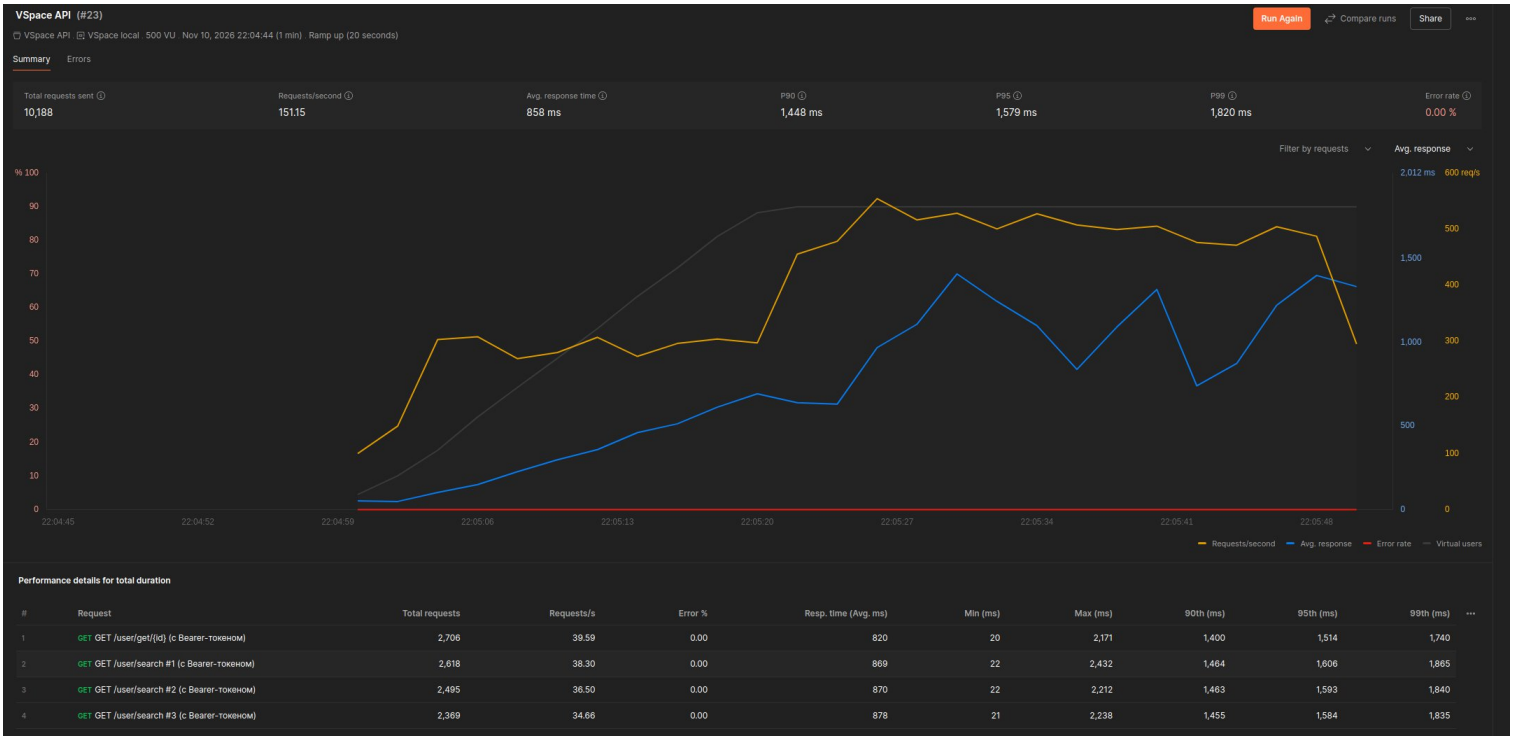
100 VU



Performance details for total duration

#	Request	Total requests	Requests/s	Error %	Resp. time (Avg. ms)	Min (ms)	Max (ms)	90th (ms)	95th (ms)	99th (ms)
1	GET GET /user/get/{id} (с Bearer-токеном)	2,744	40.91	0.00	56	19	245	94	110	147
2	GET GET /user/search #1 (с Bearer-токеном)	2,742	40.88	0.00	64	21	237	109	128	166
3	GET GET /user/search #2 (с Bearer-токеном)	2,740	40.85	0.00	64	20	241	107	127	168
4	GET GET /user/search #3 (с Bearer-токеном)	2,736	40.79	0.00	63	20	250	105	124	164

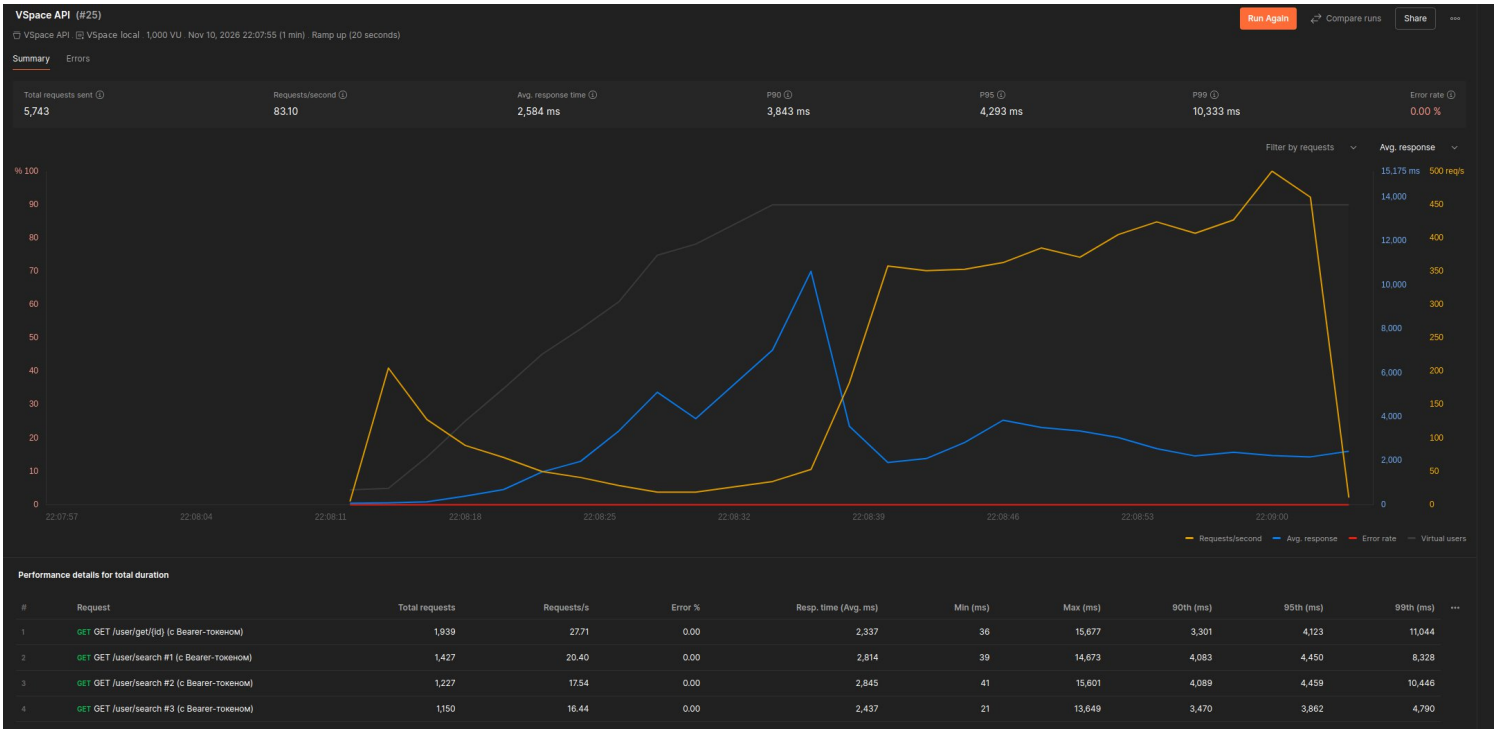
500 VU

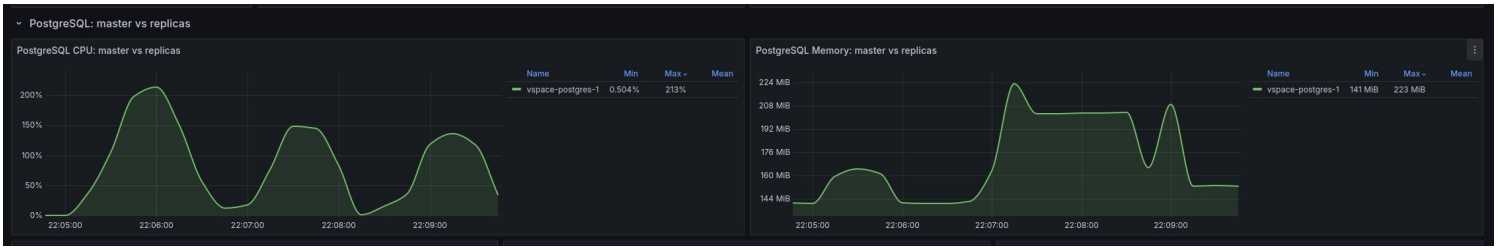


Performance details for total duration

#	Request	Total requests	Requests/s	Error %	Resp. time (Avg. ms)	Min (ms)	Max (ms)	90th (ms)	95th (ms)	99th (ms)
1	GET GET /user/get/{id} (с Bearer-токеном)	2,706	39.59	0.00	820	20	2,171	1,400	1,514	1,740
2	GET GET /user/search #1 (с Bearer-токеном)	2,618	38.30	0.00	869	22	2,432	1,464	1,606	1,865
3	GET GET /user/search #2 (с Bearer-токеном)	2,495	36.50	0.00	870	22	2,212	1,463	1,593	1,840
4	GET GET /user/search #3 (с Bearer-токеном)	2,369	34.66	0.00	878	21	2,238	1,455	1,584	1,835

1000 VU

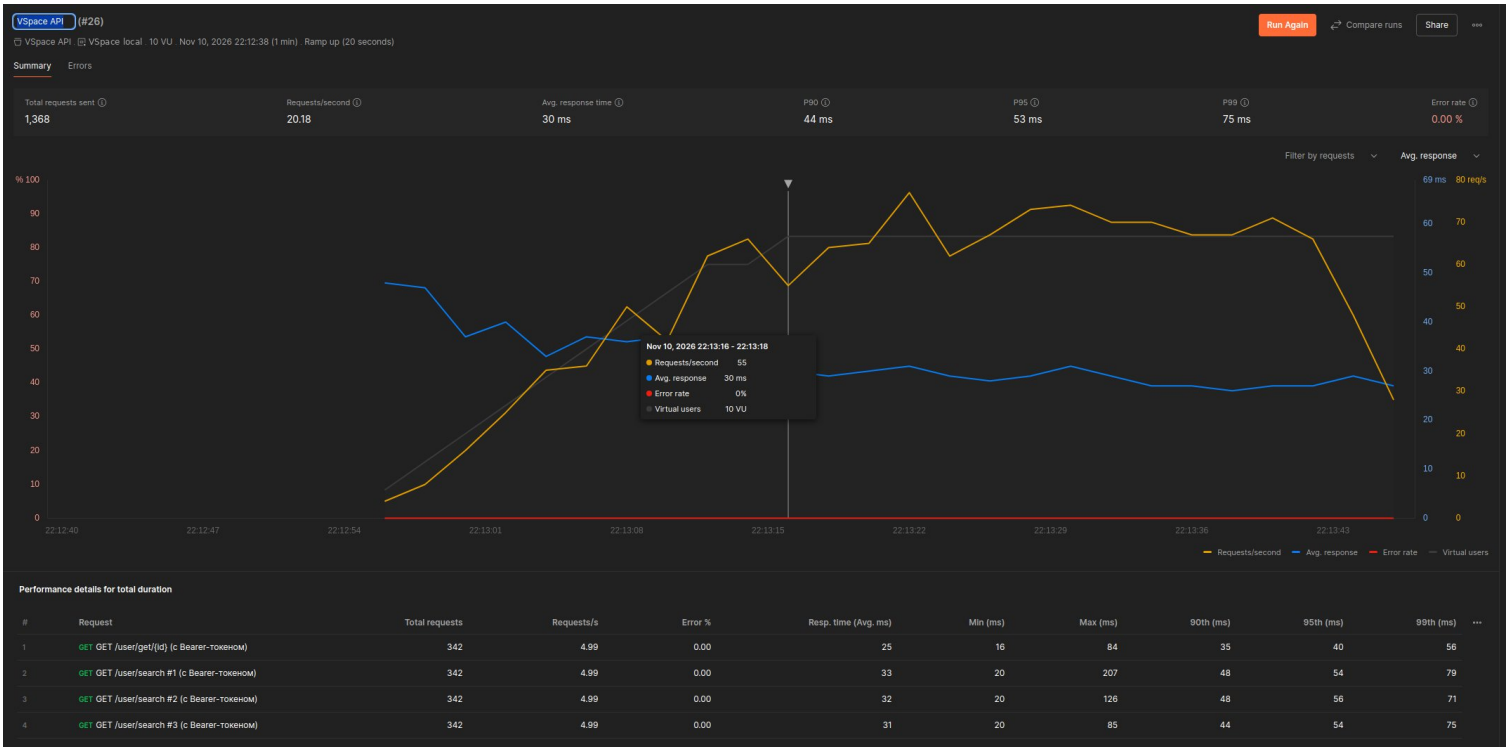

VSpace API (#25)
Run Again
Compare runs
Share
VSpace API · VSpace local · 1,000 VU · Nov 10, 2026 22:07:55 (1 min) · Ramp up (20 seconds)
Summary
Errors
Total requests sent
5,743
Requests/second
83.10
Avg. response time
2,584 ms
P90
3,843 ms
P95
4,293 ms
P99
10,333 ms
Error rate
0.00 %
Filter by requests
Avg. response
Requests/second
Avg. response
Error rate
Virtual users
Performance details for total duration
Request Total requests Requests/s Error % Resp. time (Avg. ms) Min (ms) Max (ms) 90th (ms) 95th (ms) 99th (ms)
1 GET GET /user/get/{id} (с Bearer-токеном) 1,939 27.71 0.00 2,337 36 15,677 3,301 4,123 11,044
2 GET GET /user/search #1 (с Bearer-токеном) 1,427 20.40 0.00 2,814 39 14,673 4,083 4,450 8,328
3 GET GET /user/search #2 (с Bearer-токеном) 1,227 17.54 0.00 2,845 41 15,601 4,089 4,459 10,446
4 GET GET /user/search #3 (с Bearer-токеном) 1,150 16.44 0.00 2,437 21 13,649 3,470 3,862 4,790


PostgreSQL: master vs replicas
PostgreSQL CPU: master vs replicas
Name Min Max Mean
vspace-postgres-1 0.504% 213%
PostgreSQL Memory: master vs replicas
Name Min Max Mean
vspace-postgres-1 141 MiB 223 MiB

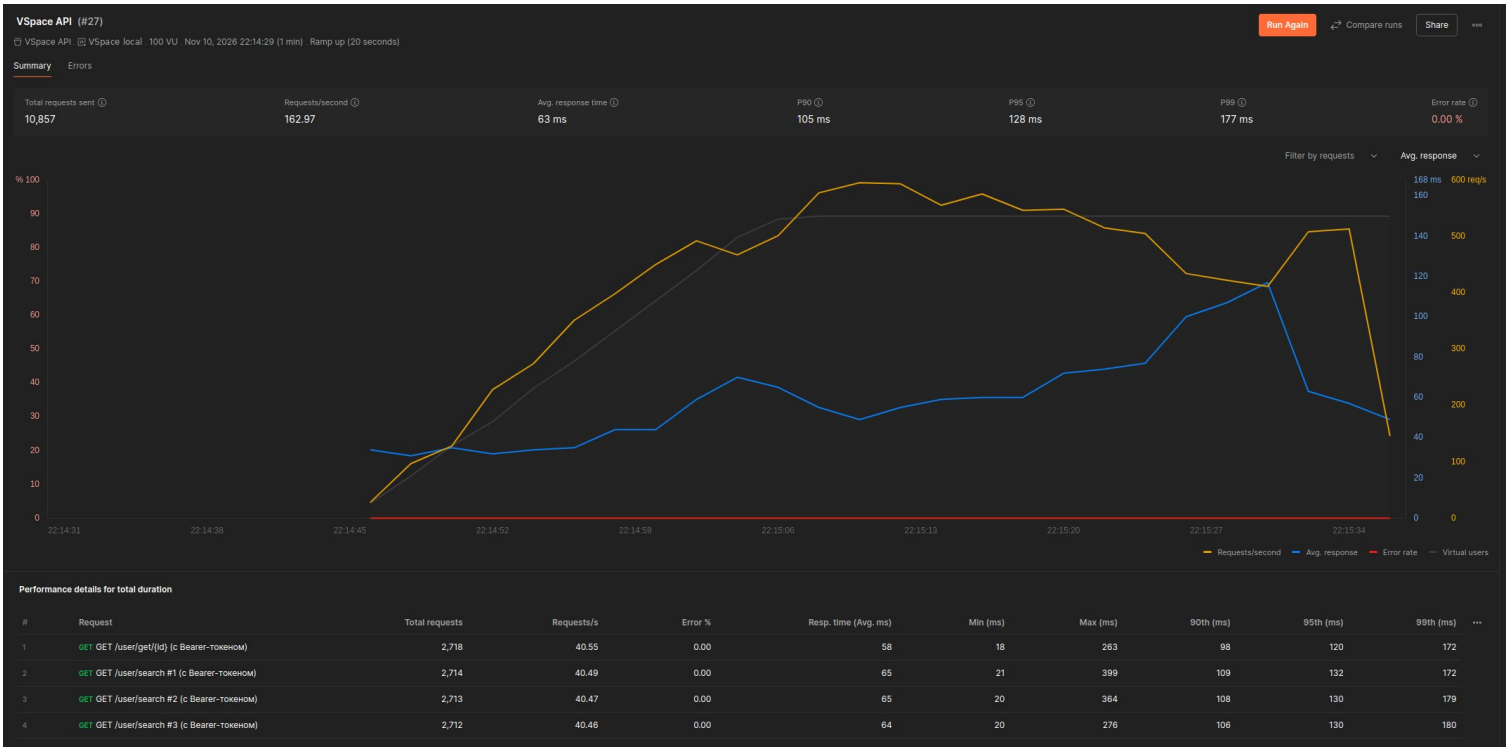
3. Синхронная потоковая реплика (1 мастер + 2 слейва)

./scripts/replication.sh up

10 VU



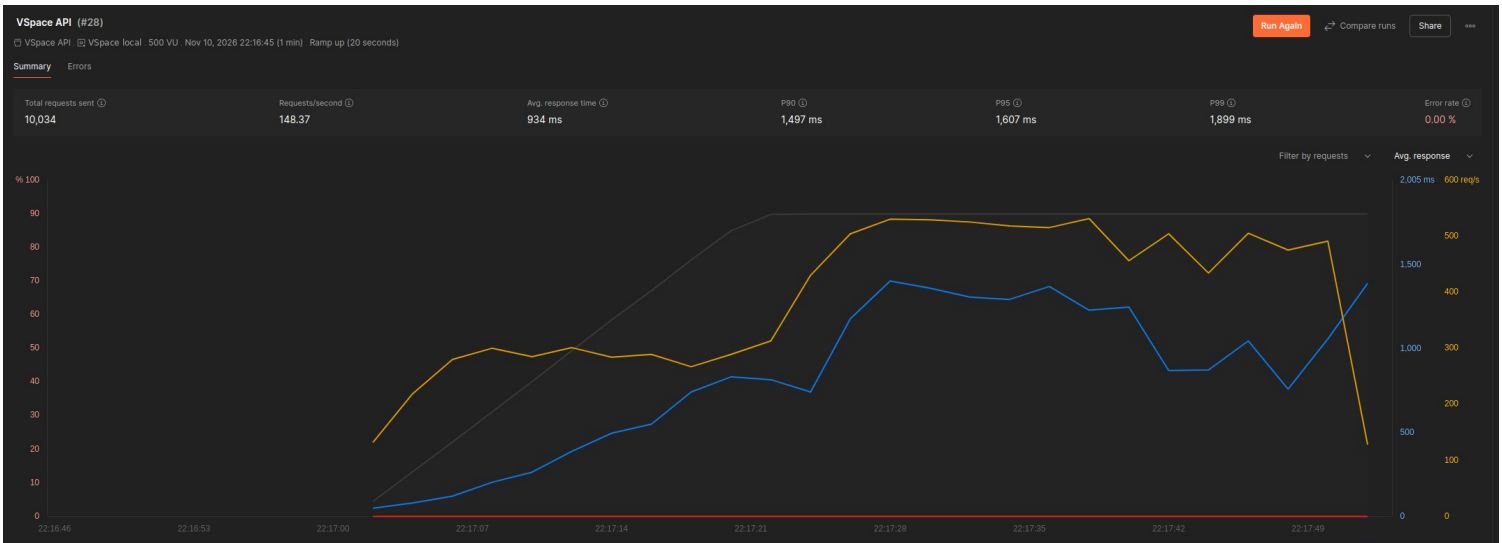
100 VU



Performance details for total duration

#	Request	Total requests	Requests/s	Error %	Resp. time (Avg. ms)	Min (ms)	Max (ms)	90th (ms)	95th (ms)	99th (ms)
1	GET GET /user/get/{id} (с Bearer-токеном)	2,718	40.55	0.00	58	18	263	98	120	172
2	GET GET /user/search #1 (с Bearer-токеном)	2,714	40.49	0.00	65	21	399	109	132	172
3	GET GET /user/search #2 (с Bearer-токеном)	2,713	40.47	0.00	65	20	364	108	130	179
4	GET GET /user/search #3 (с Bearer-токеном)	2,712	40.46	0.00	64	20	276	106	130	180

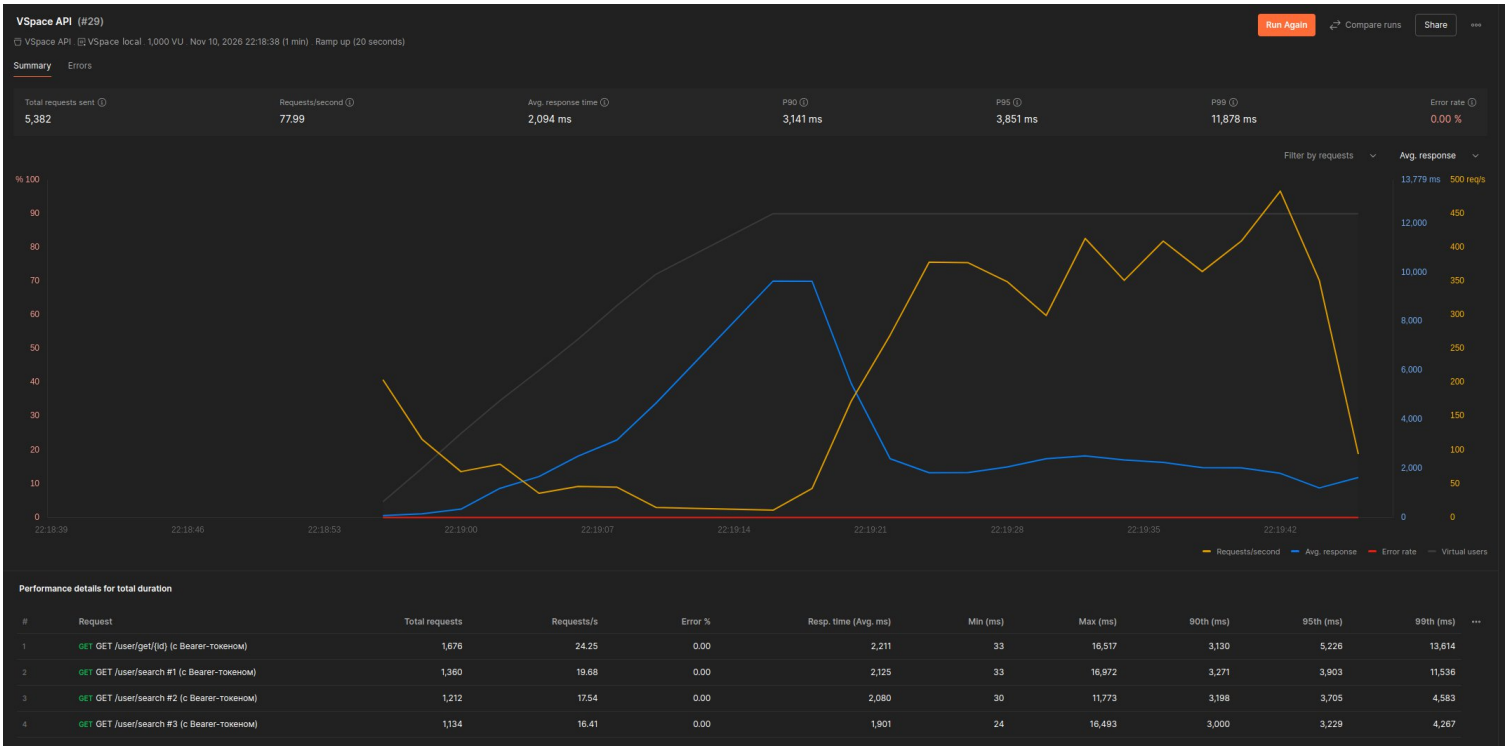
500 VU

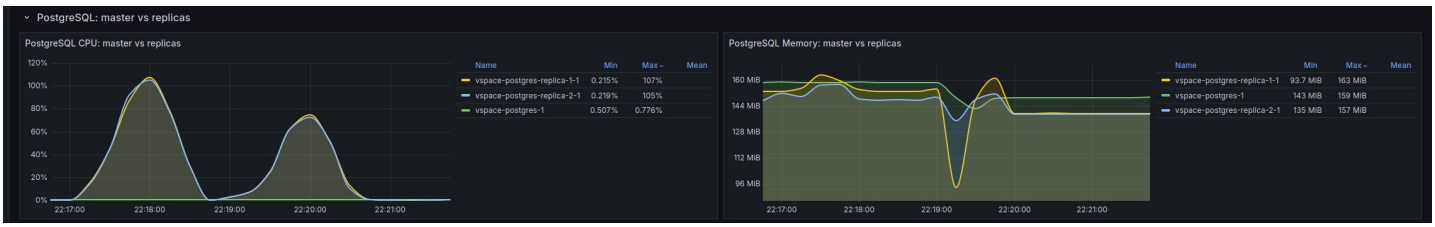


Performance details for total duration

#	Request	Total requests	Requests/s	Error %	Resp. time (Avg. ms)	Min (ms)	Max (ms)	90th (ms)	95th (ms)	99th (ms)
1	GET GET /user/get/{id} (с Bearer-токеном)	2,670	39.27	0.00	891	21	2,695	1,454	1,547	1,891
2	GET GET /user/search #1 (с Bearer-токеном)	2,552	37.53	0.00	923	25	2,577	1,509	1,623	1,908
3	GET GET /user/search #2 (с Bearer-токеном)	2,464	36.24	0.00	945	23	2,669	1,506	1,607	1,889
4	GET GET /user/search #3 (с Bearer-токеном)	2,348	34.53	0.00	980	24	2,461	1,520	1,642	1,937

1000 VU


VSpace API (#29)
VSpace API · VSpace local · 1,000 VU · Nov 10, 2026 22:18:38 (1 min) · Ramp up (20 seconds)
Run Again
Compare runs
Share
Summary
Errors
Total requests sent
5,382
Requests/second
77.99
Avg. response time
2,094 ms
P90
3,141 ms
P95
3,851 ms
P99
11,878 ms
Error rate
0.00 %
Filter by requests
Avg. response
% 100
90
80
70
60
50
40
30
20
10
0
13,779 ms
500 req/s
12,000
10,000
8,000
6,000
4,000
2,000
0
450
400
350
300
250
200
150
100
50
0
22:18:39
22:18:46
22:18:53
22:19:00
22:19:07
22:19:14
22:19:21
22:19:28
22:19:35
22:19:42
Requests/second
Avg. response
Error rate
Virtual users
Performance details for total duration
Request Total requests Requests/s Error % Resp. time (Avg. ms) Min (ms) Max (ms) 90th (ms) 95th (ms) 99th (ms)
1 GET GET /user/get/{id} (с Bearer-токеном) 1,676 24.25 0.00 2,211 33 16,517 3,130 5,226 13,614
2 GET GET /user/search #1 (с Bearer-токеном) 1,360 19.68 0.00 2,125 33 16,972 3,271 3,903 11,536
3 GET GET /user/search #2 (с Bearer-токеном) 1,212 17.54 0.00 2,080 30 11,773 3,198 3,705 4,583
4 GET GET /user/search #3 (с Bearer-токеном) 1,134 16.41 0.00 1,901 24 16,493 3,000 3,229 4,267


PostgreSQL: master vs replicas
PostgreSQL CPU: master vs replicas
120%
100%
80%
60%
40%
20%
0%
22:17:00
22:18:00
22:19:00
22:20:00
22:21:00
Name Min Max Mean
vspace-postgres-replica-1-1 0.215% 107%
vspace-postgres-replica-2-1 0.219% 105%
vspace-postgres-1 0.507% 0.776%
PostgreSQL Memory: master vs replicas
160 MiB
144 MiB
128 MiB
112 MiB
96 MiB
22:17:00
22:18:00
22:19:00
22:20:00
22:21:00
Name Min Max Mean
vspace-postgres-replica-1-1 93.7 MiB 163 MiB
vspace-postgres-1 143 MiB 159 MiB
vspace-postgres-replica-2-1 135 MiB 157 MiB

РЕЗУЛЬТАТЫ СРАВНЕНИЯ: МАСТЕР И РЕПЛИКИ

VSpace, PostgreSQL 17, 16 vCPU / 30 ГБ / NVMe, все контейнеры на одном хосте.

Асинхронная репликация, Read/write split: SELECT на реплики, запись на мастер.

ИСХОДНЫЕ ДАННЫЕ

8 прогонов Postman по 1 мин (ramp-up 20 с), только чтение: GET /user/get + 3 x GET /user/search.

1-4 (runs #21-23, #25) - мастер, 5-8 (runs #26-29) - реплики. Ошибок 0% везде.

СВОДНАЯ ТАБЛИЦА (мастер -> реплики)

VU	RPS	Avg, мс	P95, мс	P99, мс
10	20.59 -> 20.18 (-2%)	28 -> 30 (+7%)	49 -> 53 (+8%)	68 -> 75 (+10%)
100	164.59 -> 162.97 (-1%)	62 -> 63 (+2%)	122 -> 128 (+5%)	164 -> 177 (+8%)
500	151.15 -> 148.37 (-2%)	858 -> 934 (+9%)	1579 -> 1607 (+2%)	1820 -> 1899 (+4%)
1000	83.10 -> 77.99 (-6%)	2584 -> 2094 (-19%)	4293 -> 3851 (-10%)	10333 -> 11878 (+15%)

ПО ЭНДПОИНТАМ (среднее, мс, мастер -> реплики)

VU	GET /user/get	search #1	search #2	search #3
100	56 -> 58	64 -> 65	64 -> 65	63 -> 64
500	820 -> 891	869 -> 923	870 -> 945	878 -> 980
1000	2337 -> 2211	2814 -> 2125	2845 -> 2080	2437 -> 1901

РАСПРЕДЕЛЕНИЕ РЕСУРСОВ

CPU, 2000 запросов	PHP	Мастер	Реплика 1	Реплика 2	Итого БД

Чтение с мастера	~560%	~431%	0%	0%	~431%
Чтение с реплик	~570%	3%	205%	210%	~415%

ВЛИЯНИЕ НАСТРОЕК (c=1, /user/search)

Вариант	p50 /user/search
Xdebug debug, напрямую	33.5 мс
Xdebug off, напрямую	26.2 мс
Xdebug off + PgBouncer	21.1 мс

ВЫВОДЫ

1) Замеры относятся к read-пути: все прогоны только читающие.

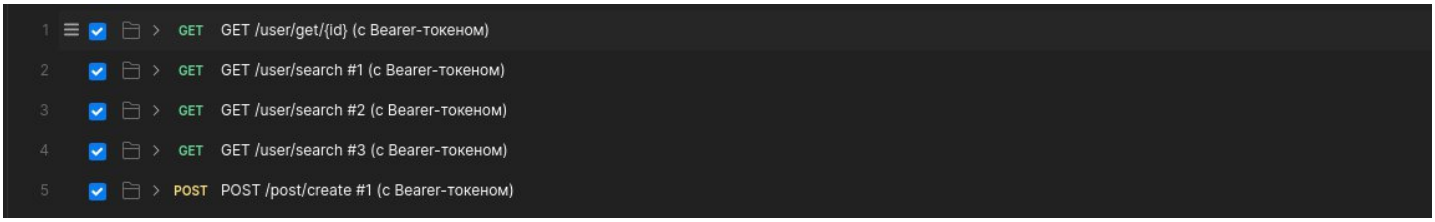
2) 10-100 VU - паритет; 500 VU - реплики хуже на 2% по RPS и 9% по avg; 1000 VU - на 6% по RPS и 15% по p99.

3) Суммарная работа БД не меняется: 431% на мастере против 205% + 210% на двух репликах, то есть те же ~420%, просто поделённые между процессами.

4) Причина - общая инфраструктура: одни 16 ядер, один NVMe, общий page cache; реплика дополнительно тратит CPU на replay WAL. Новых ресурсов нет, только накладные расходы.

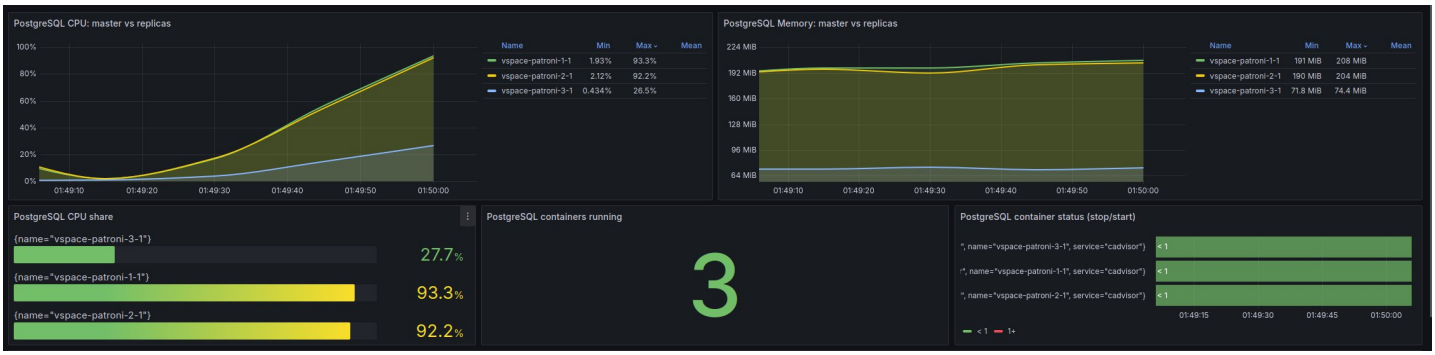
5) На одном хосте реплики не масштабируют чтение; прирост дают меры, снижающие работу: PgBouncer, XDEBUG_MODE=off и сокращение объёма ответа.

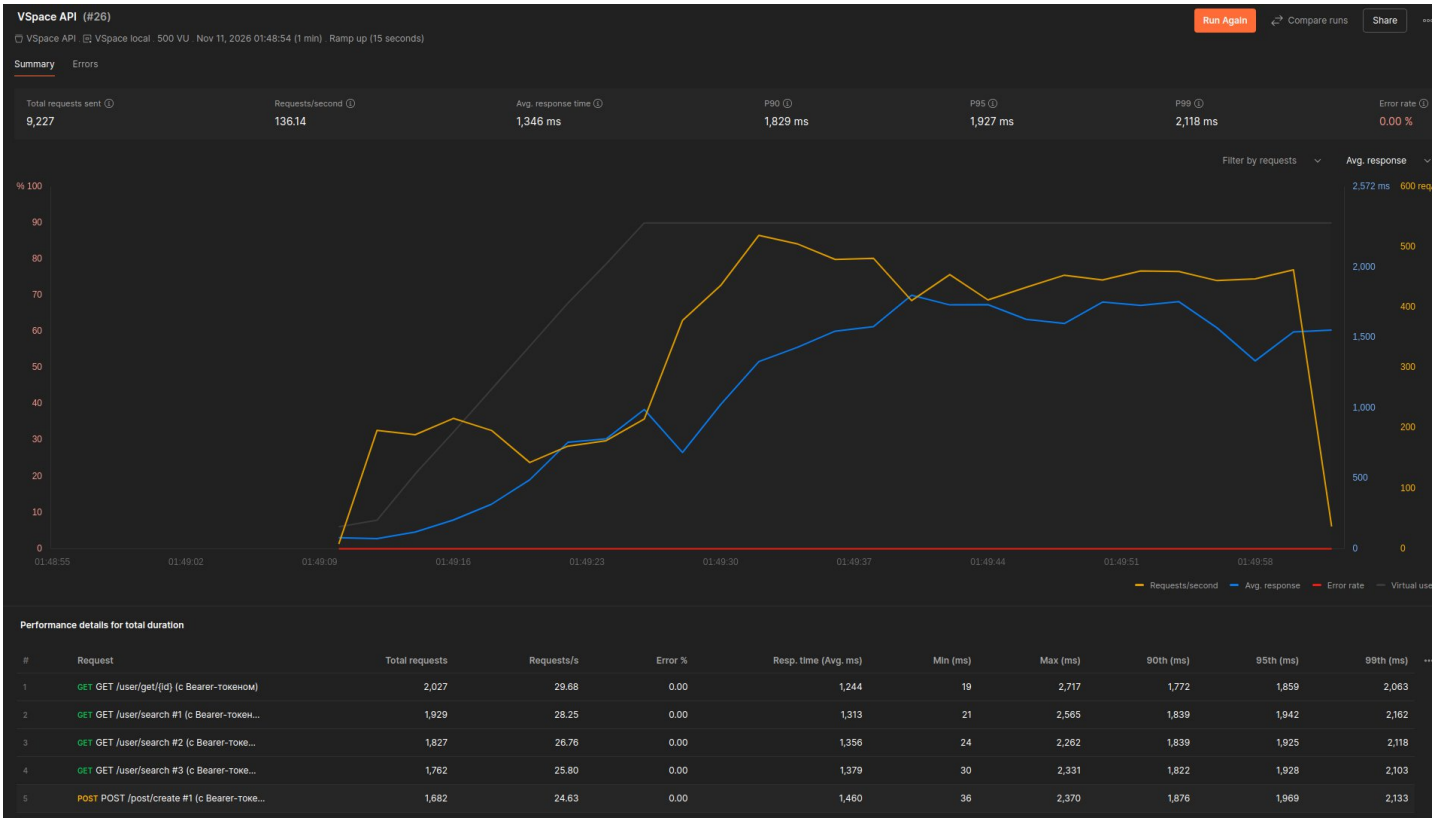
4. Кворумная синхронная репликация с нагрузкой на запись + чтение



./scripts/patroni.sh up

./scripts/patroni.sh use



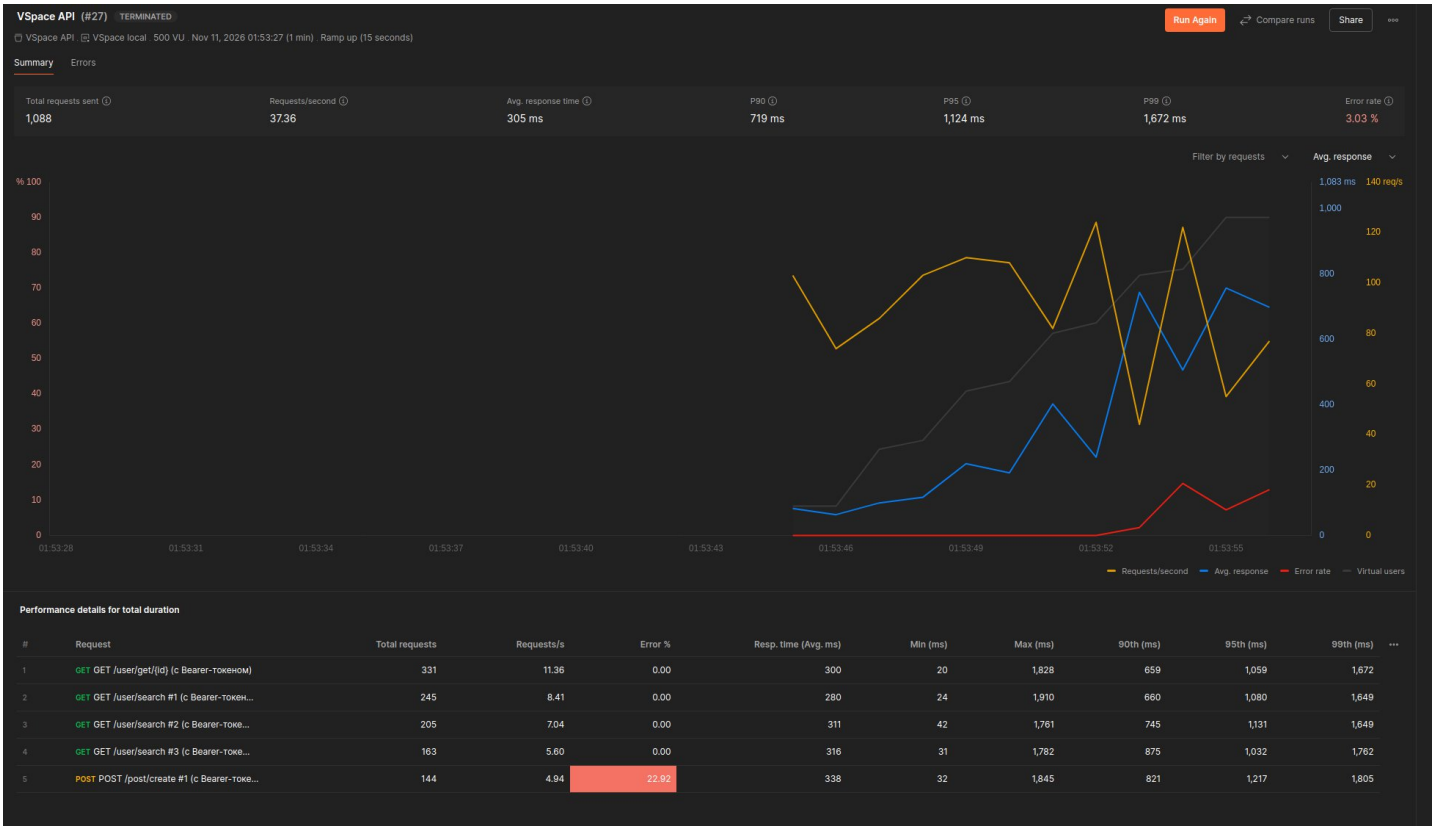


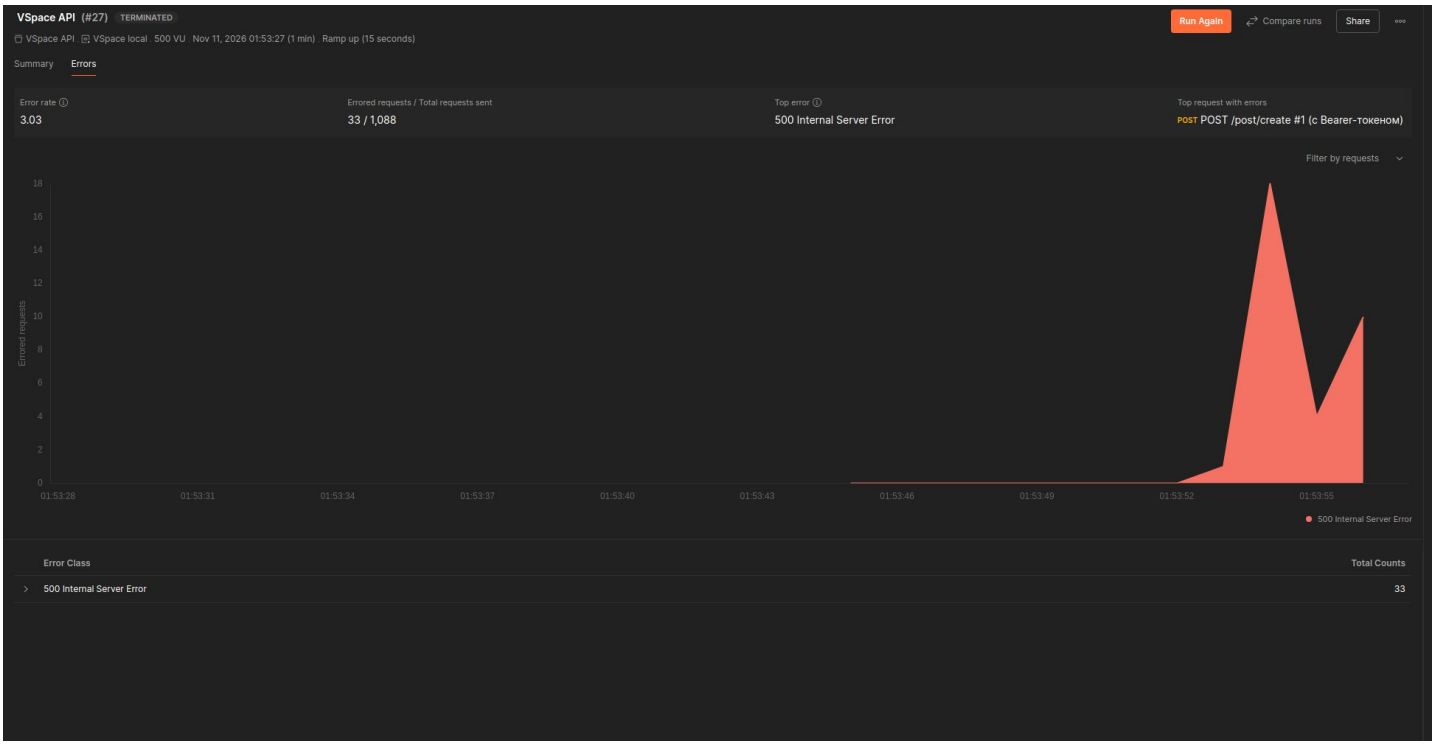
5. Остановка мастера, перенос мастера на актуальный слейв

./scripts/patroni.sh use-legacy

./scripts/patroni.sh down

./scripts/replication.sh up





./scripts/failover-drill.sh drill --manual --yes --user 6f61227e-3efe-4874-91c8-65f265845f6f

Postman остановлен. Enter — снять baseline...

baseline постов пользователя 6f61227e-3efe-4874-91c8-65f265845f6f: 0

Запустите Postman, дайте нагрузке разогреться. Enter — убить мастер...

убиваем мастер (01:53:53)

Остановите Postman. Enter — продолжить...

Сколько POST /post/create вернуло 200? 111

успешных create по отчёту Postman: 111

  postgres-replica-1: replay_lsn=1/581774D0

  postgres-replica-2: replay_lsn=1/581774D0

новый мастер: postgres-replica-1, второй слейв: postgres-replica-2

промоут за 110,41c, replay lag: 110.509c

второй слейв подключился к новому мастеру: true

================ РЕЗУЛЬТАТ ================

отправлено успешно (клиент): 111

записано в БД: 111

потеряно транзакций: 0

replay lag на промоуте: 110.509c

отчёт: ht/manual_restore_master/20261111-015318/report.json

==

{

  "run": "20261111-015318",

  "mode": "manual",

  "master_killed": "postgres",

  "promoted": "postgres-replica-1",

  "repointed": "postgres-replica-2",

  "sent_ok": 111,

  "written": 111,

  "lost_transactions": 0,

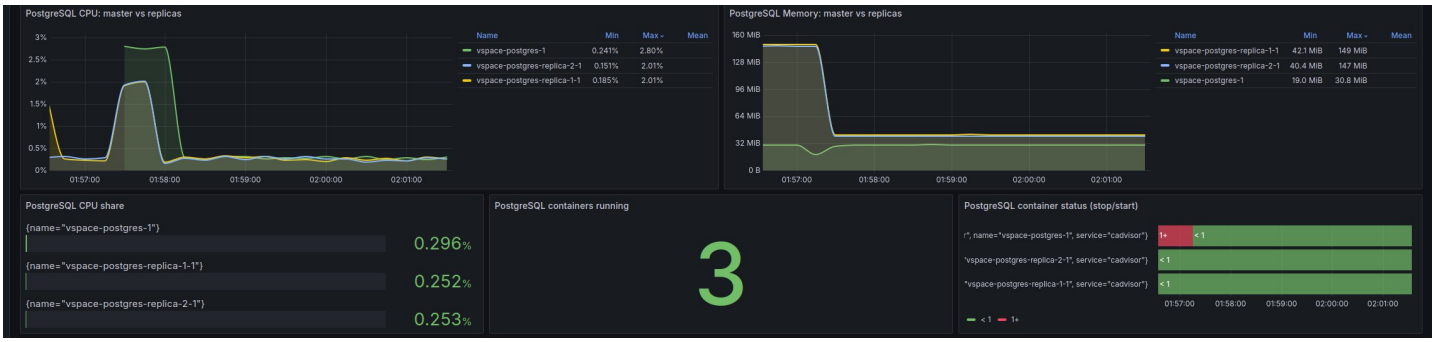
  "replica_rejoined": true,

  "replay_lag_seconds_at_promotion": 110.509,

  "promotion_seconds": 110,415

}

Откат: scripts/failover-drill.sh restore --yes



public.posts: 111 rows total

Next Show all Sortin

#	1 id	2 author_user_id	3 text	4 created_at	5 updated_at
60	01a1c7aa-e6eb-72d0-bd6c-62e81b6...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:48	2026-11-10 22:53:48
61	01a1c7aa-e6eb-72be-94ac-5f255e5f...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:48	2026-11-10 22:53:48
62	01a1c7aa-e6bf-7059-a28c-d2ecbedc...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:48	2026-11-10 22:53:48
63	01a1c7aa-e6ad-711f-be6e-efeedb2c...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:48	2026-11-10 22:53:48
64	01a1c7aa-e699-7353-9b49-59551b3...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:48	2026-11-10 22:53:48
65	01a1c7aa-e680-7185-bf40-e91a7bea...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:48	2026-11-10 22:53:48
66	01a1c7aa-e5db-72e3-a4ba-f885257a...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:48	2026-11-10 22:53:48
67	01a1c7aa-e531-73ff-808f-b62954996...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:48	2026-11-10 22:53:48
68	01a1c7aa-e510-7302-b626-add6155...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:48	2026-11-10 22:53:48
69	01a1c7aa-e4d0-7391-9716-7de29d3...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:48	2026-11-10 22:53:48
70	01a1c7aa-e4be-7366-94cc-0816d944...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:48	2026-11-10 22:53:48
71	01a1c7aa-e49d-7390-9e59-b3d8427...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:48	2026-11-10 22:53:48
72	01a1c7aa-e42f-71b6-a0ff-9f883f158...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:47	2026-11-10 22:53:47
73	01a1c7aa-e3f6-704b-a416-6557c395...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:47	2026-11-10 22:53:47
74	01a1c7aa-e3d5-72f9-8409-8d27d917...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:47	2026-11-10 22:53:47
75	01a1c7aa-e3ca-7030-b8d8-272108a5...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:47	2026-11-10 22:53:47
76	01a1c7aa-e204-7331-8b9a-c8fafb86...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:47	2026-11-10 22:53:47
77	01a1c7aa-e1bf-7127-abaa-1f546b50...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:47	2026-11-10 22:53:47
78	01a1c7aa-ddc7-7389-9a0a-a4577a10...	6f61227e-3efe-4874-91c8-65f265845...	1	2026-11-10 22:53:46	2026-11-10 22:53:46
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